

Week 1 – Understanding the Business Problem

Requirement Analysis Document – E-Commerce Management System

1. Introduction

An online shopping platform such as Amazon or Flipkart allows customers to purchase products through a website or mobile application. Customers can register, browse products, add items to a cart, place orders, make payments, and track shipments.

The proposed system is an **E-Commerce Management System** that manages customers, products, categories, orders, payments, sellers, and shipments efficiently.

2. Problem Statement

Traditional shopping requires customers to physically visit stores to search for products and make purchases. An online shopping system solves this problem by providing a digital platform where customers can:

- Create an account
- Search and browse products
- View product details and prices
- Add products to a shopping cart
- Place orders
- Make online payments
- Track shipments
- View previous orders

The system also helps sellers and administrators manage products, inventory, orders, and customers.

3. Major Activities in an E-Commerce System

Customer Activities

1. Customer registration
2. Customer login
3. Browse products
4. Search and filter products
5. View product details
6. Add products to cart
7. Update or remove cart items
8. Place an order
9. Make payment
10. Track shipment
11. View order history
12. Give product reviews and ratings

Seller Activities

1. Seller registration/login
2. Add products
3. Update product information
4. Manage product stock
5. View customer orders

6. Update order status
7. Manage shipments

Administrator Activities

1. Manage customers
2. Manage sellers
3. Manage products
4. Manage categories
5. Monitor orders
6. Manage payments
7. Generate reports

4. Functional Requirements

Requirement	Description
User Registration	Customers can create a new account.
User Login	Registered users can securely log in.
Product Management	Products can be added, updated, and removed.
Product Search	Customers can search for products.
Product Categories	Products can be organized into categories.
Shopping Cart	Customers can add, remove, and update products in their cart.
Order Management	Customers can place and view orders.
Payment Processing	The system records and processes payments.
Shipment Tracking	Customers can track their order shipment status.
Reviews & Ratings	Customers can review purchased products.
Inventory Management	Sellers can monitor available product stock.
Reports	Admin can generate sales, inventory, and order reports.

5. Non-Functional Requirements

- **Performance:** The system should respond quickly to user requests.
- **Security:** Customer and payment information should be protected.
- **Availability:** The system should be available whenever customers need it.
- **Scalability:** The system should support an increasing number of customers and products.
- **Usability:** The interface should be simple and easy to understand.
- **Reliability:** Orders and payment information should be stored accurately.
- **Maintainability:** The system should be easy to update and maintain.

6. Main Entities

- Customer
- Seller
- Product
- Category
- Cart
- Cart Item
- Order

- Order Item
- Payment
- Shipment
- Review

Basic Relationships

Customer → Cart → Cart Items → Products

Customer → Order → Order Items → Products

Order → Payment

Order → Shipment

Product → Category

Seller → Products

7. Expected System Workflow

Customer Registration → Login → Browse/Search Products → View Product Details → Add to Cart → Checkout → Place Order → Payment Processing → Order Confirmation → Shipment → Shipment Tracking → Order Delivered

8. Conclusion

The proposed **E-Commerce Management System** provides a complete platform for online shopping. It connects customers, sellers, and administrators and manages the complete process from product browsing to order delivery. The requirement analysis identified the major functionalities and entities that will be used in the upcoming stages of the project, particularly database design and implementation.